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 Sección: 20 prof: Pablo Koch

Laboratorio 3

Problema 2:

$$\begin{array}{lll}
 q_0 = \epsilon & q_5 = q_2 \epsilon + q_4 \epsilon & q_{10} = q_9 \epsilon \\
 q_1 = q_0 \epsilon & q_6 = q_5 \epsilon & q_{11} = q_{10} \epsilon + q_7 \epsilon \\
 q_2 = q_1 0 & q_7 = q_6 \epsilon & q_{12} = q_5 \epsilon \\
 q_3 = q_0 \epsilon & q_8 = q_7 1 & q_{13} = q_{12} \epsilon \\
 q_4 = q_3 \epsilon & q_9 = q_6 \epsilon & q_{14} = q_{13} \epsilon + q_{11} \epsilon
 \end{array}$$

$$\begin{array}{l}
 q_{15} = q_{14} \epsilon + q_{16} \epsilon \\
 q_{16} = q_{15} 0 \\
 q_{17} = q_{16} \epsilon + q_{14} \epsilon
 \end{array}$$

$$\begin{array}{ll}
 q_0 = \epsilon & q_5 = 0\epsilon + \epsilon = \epsilon(0 + \epsilon) \\
 q_1 = \epsilon \cdot \epsilon = \epsilon & q_6 = (0 + \epsilon)\epsilon = (0 + \epsilon) \\
 q_2 = \epsilon \cdot 0 = 0 & q_7 = (0 + \epsilon)\epsilon = (0 + \epsilon) \\
 q_3 = \epsilon \cdot \epsilon = \epsilon & q_8 = (0 + \epsilon)1 \\
 q_4 = \epsilon \cdot \epsilon = \epsilon & q_9 = (0 + \epsilon)\epsilon = (0 + \epsilon)
 \end{array}$$

$$\begin{array}{l}
 q_{10} = (0 + \epsilon)\epsilon = (0 + \epsilon) \\
 q_{11} = (0 + \epsilon) + ((0 + \epsilon)1)\epsilon = (0 + \epsilon)(\epsilon + 1) \\
 q_{12} = (0 + \epsilon)\epsilon = (0 + \epsilon) \\
 q_{13} = (0 + \epsilon)\epsilon = (0 + \epsilon) \\
 q_{14} = (0 + \epsilon)\epsilon + ((0 + \epsilon)(\epsilon + 1))\epsilon = (0 + \epsilon)(\epsilon + (\epsilon + 1)) \\
 = (0 + \epsilon)(\epsilon + 1) \quad q_5 \\
 q_{15} = (0 + \epsilon)(\epsilon + 1) + q_{16}\epsilon = (0 + \epsilon)(\epsilon + 1) + q_{16}
 \end{array}$$

$$q_{15} = (0 + \epsilon)(\epsilon + 1) + q_{15}0 \quad \text{pos} \quad q_{16} = q_{15}0$$

R Q R P

$$q_{15} = (0 + \epsilon)(\epsilon + 1) 0^*$$

$$q_{16} = (0 + \epsilon)(\epsilon + 1) 0^* 0$$

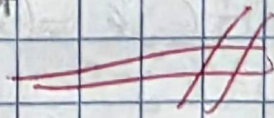
$$q_{17} = (0 + \epsilon)(\epsilon + 1) 0^* 0 + (0 + \epsilon)(\epsilon + 1) \\ = (0 + \epsilon)(\epsilon + 1)(0^* 0 + \epsilon)$$

$$0^* 0 + \epsilon = 0^*$$

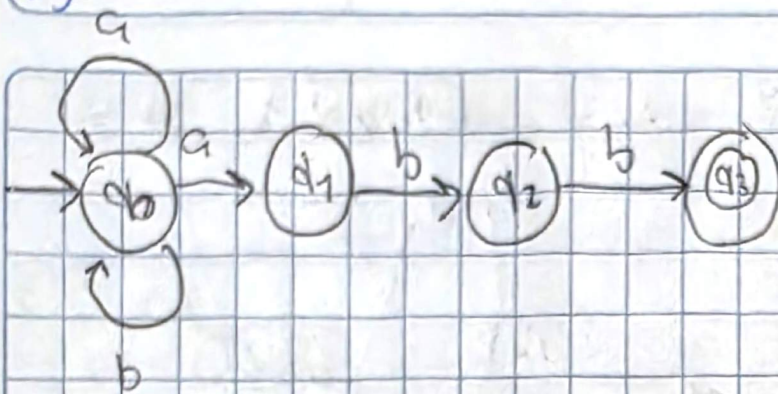
$0^+ + \epsilon = (0^+ \epsilon)$

$$q_{17} = (0 + \epsilon)(\epsilon + 1) 0^*$$

$$R_F = (0 + \epsilon)(\epsilon + 1) 0^* \\ = (0 | \epsilon)(\epsilon | 1) 0^* \\ = 0?1?0^*$$



1)



$$q_0 = \epsilon + q_0 a + q_0 b$$

$$q_0 = \epsilon + q_0 (a + b)$$

$$q_0 = \epsilon (a + b)^*$$

$$q_0 = (a + b)^*$$

$$q_1 = q_0 a$$

$$q_1 = (a + b)^* a$$

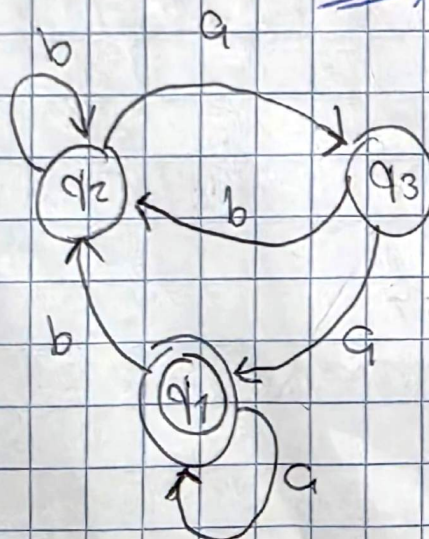
$$q_2 = q_1 b$$

$$q_2 = (a + b)^* a b$$

$$q_3 = q_2 b$$

$$q_3 = (a + b)^* a b b$$

2.)



$$q_1 = q_1 b + q_1 a + q_3 a$$

$$q_3 = q_2 a + q_1$$

$$q_2 = q_1 b + q_2 b + q_3 b$$

$$q_2 b = q_2 - q_2 b - q_3 b$$

$$q_2 b = (q_1 b + q_2 b + (q_2 a) b)$$

$$q_2 = \downarrow$$

$$q_1 b + q_2 b + (q_2 a) b$$

$$q_1 b + q_2 (b + a b)$$

$$q_2 = q$$

$$q_2 = q_1 b \cdot (b + a b)^*$$

$$q_2 = q_1 b \cdot (b + ab)^*$$

$$q_2 = q_2 a + q_2 a a$$

$$q_2 = q_1 a + (b \cdot (b + ab)^*) a a$$

$$q_2 = (b \cdot (b + ab)^*) a a \cdot q_1^*$$

$$q_1 = \underbrace{a_1 a}_{R \ P} + \underbrace{B}_{R}$$

$$q_1 = B a^*$$

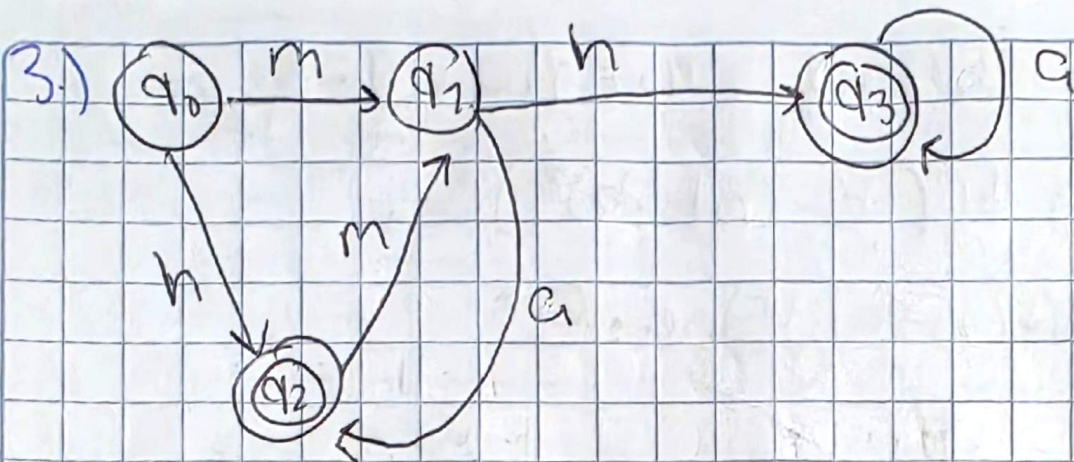
$$q_2 = (b \cdot (b + ab)^*) a a \cdot q_1^*$$

$$q_1 = q_1 a + (q_2 a) a$$

$$q_1 = q_1 a + (q_1 b \cdot (b + ab)^* a) a$$

$$q_1 = q_1 (a + b \cdot (b + ab)^* a a) + \epsilon$$

$$q_1 = (a + b (b + ab)^* a a)^*$$



$$q_0 = \epsilon$$

$$q_1 = q_0 m + q_2 m$$

$$q_2 = q_0 h + q_1 a$$

$$q_1 = m + q_2 m$$

$$q_1 = m + (h + q_1 a) m$$

$$q_1 = m + h m + q_1 a m$$

$$q_1 = (m + h m) \cdot (a m)^* \rightarrow q_1$$

q1

$$q_2 = q_0 h + q_1 a$$

$$q_2 = q_0 h + (q_0 m + q_2 m) a$$

$$q_2 = h + a m + q_2 a m$$

$$q_2 = (h + a m) (a m)^*$$

$$R_F = (h + a m) (a m)^* (m + h m) (a m)^* h a^*$$